

Discrete Mathematics Assignment - 2

Name : Harsh Hemalkumar Mehta

Student Id : 202412048

202412048

Assignment - 2 | Harsh Mehta

① • Statement 1 $\exists x, \forall y, x-y=5$ where $x \in \mathbb{R}$ & $y \in \mathbb{R}$.

$$x = 7 \text{ Fixed } y = 2$$

$$\therefore x - y = 7 - 2 = 5$$

$$\text{take } x = 7 \quad y = 3$$

$$\therefore x - y = 7 - 3 = 4$$

$\therefore$ It is not true for every statement of y .

• Statement 2: $\forall x, \exists y, x-y=5$ where $x \in \mathbb{R}, y \in \mathbb{R}$.

$$\text{let } x = a \in \mathbb{R} \text{ and } y = b \in \mathbb{R}$$

$$\therefore x - y = a - b = 5$$

$$a - b = 5$$

$$b = a - 5$$

$$\therefore \forall x \exists y = x - 5 \in \mathbb{R} \quad \exists \cdot x - 5 = 5$$

where $x \in \mathbb{R}$ and $y \in \mathbb{R}$.

$\therefore$ Statement 2 is correct.

Q-2 $\forall x [P(x) \vee Q(x)] \Rightarrow [\forall x P(x) \vee \forall x Q(x)]$
Prove or Disprove.

Ans

Suppose that $\forall x (P(x) \vee Q(x))$ is true
it means ^{at least} ~~at least~~ from the domain,
then $P(a) \vee Q(a)$ is true.

$\therefore$ Either $P(a)$ is true or $Q(a)$ is true

$\therefore$ We can not prove that $\forall x P(x) \vee \forall x Q(x)$
is true because same value of x is
true not every value.

eg $P(x) : x+1$ is odd and $Q(x) = x+1$ is even
Domain $\subseteq \mathbb{N}$

$\forall x (P(x) \vee Q(x))$

$\rightarrow x = 1$

$P(x) = 2$ is ~~odd~~ _x or $Q(x) = 2$ is even. _x

$\rightarrow x = 2$

$P(x) = 3$ is odd. _x or $Q(x) = 3$ is even. _x

$\therefore$ We can ~~pro~~ disprove that

$\forall x [P(x) \vee Q(x)] \Rightarrow [\forall x P(x) \vee \forall x Q(x)]$

③ Prove that if n is an integer and $n^3 + 5$ is odd then n is even.

A $\therefore$ Using Contra Positive rule.

If " $n^3 + 5$ is odd n is even" is false then given statement is true.
 $\therefore$ means n is odd

Then by the definition of odd integer.
 $n = 2k + 1$

by substituting $2k + 1$ for n .
we find.

$$\begin{aligned} (2k+1)^3 + 5 &= 8k^3 + 12k^2 + 6k + 6 \\ &= 2(4k^3 + 6k^2 + 3k + 3) \\ \therefore \text{It is even.} \end{aligned}$$

$\therefore$ The negation of $(2k+1)^3 + 5$ is even there for its false.

So from this we can say that our conditional statement is proven.

Q-4 Prove Sum of two even integers is also even.

A-4 Proof: Let a & b are two even integers

$$\therefore a = 2k \quad b = 2m \text{ for some } k, m$$

$$\therefore a + b = 2k + 2m \\ = 2(k + m)$$

$\therefore$ So, it is the multiple of 2 therefore its even.

Q-5 $A = \{1, 2, 5, 6, 7, 10, 11, 12\}$
 $B = \{x \mid x \text{ is an odd integer and } 3 \leq x \leq 10\}$
 $C = \{a, b, c, 4, 5, 6\}$

(a)

$7 \in A, 6 \notin B$

$10 \notin B$

$4 \notin A$

$4 \in C$

$4 \notin B$

$\{2, 5, 12\} \subseteq A$

$\{3, 5, 7\} \subseteq B$

$\{a, b, c, 5\} \subseteq C$

⑥①

$$\rightarrow A \cup B = \{1, 2, 3, 5, 6, 7, 10, 11, 12, 9\}$$

$$\rightarrow A \cap B = \{5, 7\}$$

$$\rightarrow A \setminus B = \{1, 2, 6, 10, 11, 12\}$$

$$\rightarrow P(A \cap B) = \{\emptyset, \{5\}, \{7\}, \{5, 7\}\}$$

$$\rightarrow (A \cap B) \times \{9, 4\} = \{(5, 9), (5, 4), (7, 9), (7, 4)\}$$

②

$$\rightarrow A \cup (B \cap C) = \{1, 2, 5, 6, 7, 10, 11, 12\}$$

$$\rightarrow A \cap (B \cup C) = \{5, 6, 7\}$$

$$\rightarrow A \setminus (B \cup C) = \{1, 2, 10, 11, 12\}$$

$$\rightarrow (A \setminus B) \setminus C = \{1, 2, 10, 11, 12\}$$

③ $\rightarrow C(A \cup B) = 10$

$$\rightarrow C(A \cap B) = 2$$

$$\rightarrow C(A \setminus B) = 6$$

$$\rightarrow C(P(A \cap B)) = 4$$

$$\rightarrow C(A \cap B \times \{9, 4\}) = 4$$

$$\rightarrow C(A \cup (B \cap C)) = 8$$

$$\rightarrow C(A \cap (B \cup C)) = 3$$

$$\rightarrow C(A \setminus (B \cup C)) = 5$$

$$\rightarrow C((A \setminus B) \setminus C) = 5$$

Q-6 Find A & B if $A \setminus B = \{1, 5, 7, 8\}$, $B \setminus A = \{2, 10\}$
 $A \cap B = \{3, 6, 9\}$

A-6 By Definition.

$$A \setminus B = \{x | x \in A \text{ \& } x \notin B\} = \{1, 5, 7, 8\}$$

$$\therefore 1, 5, 7, 8 \in A \text{ but } 1, 5, 7, 8 \notin B$$

$$B \setminus A = \{x | x \in B \text{ \& } x \notin A\} = \{2, 10\}$$

$$\therefore 2, 10 \in B \text{ but } 2, 10 \notin A$$

$$A \cap B = \{3, 6, 9\}$$

$$\therefore 3, 6, 9 \in A, B$$

$$\therefore A = \{1, 3, 5, 6, 7, 8, 9\} \text{ \& } B = \{2, 3, 6, 9, 10\}$$

Q-7 Let $A = \{a, b, c\}$, $B = \{x, y\}$, $C = \{0, 1\}$ find

(a) $C \times A \times B = \{(0, a, x), (0, a, y), (0, b, x), (0, b, y),$
 $(0, c, x), (0, c, y), (1, a, x), (1, a, y),$
 $(1, b, x), (1, b, y), (1, c, x), (1, c, y)\}$

(b) $B \times C \times A = \{(x, 0, a), (x, 0, b), (x, 0, c), (x, 1, a),$
 $(x, 1, b), (x, 1, c), (y, 0, a), (y, 0, b),$
 $(y, 0, c), (y, 1, a), (y, 1, b), (y, 1, c)\}$

Q 1-7 are attached above [End]